# Research Outline


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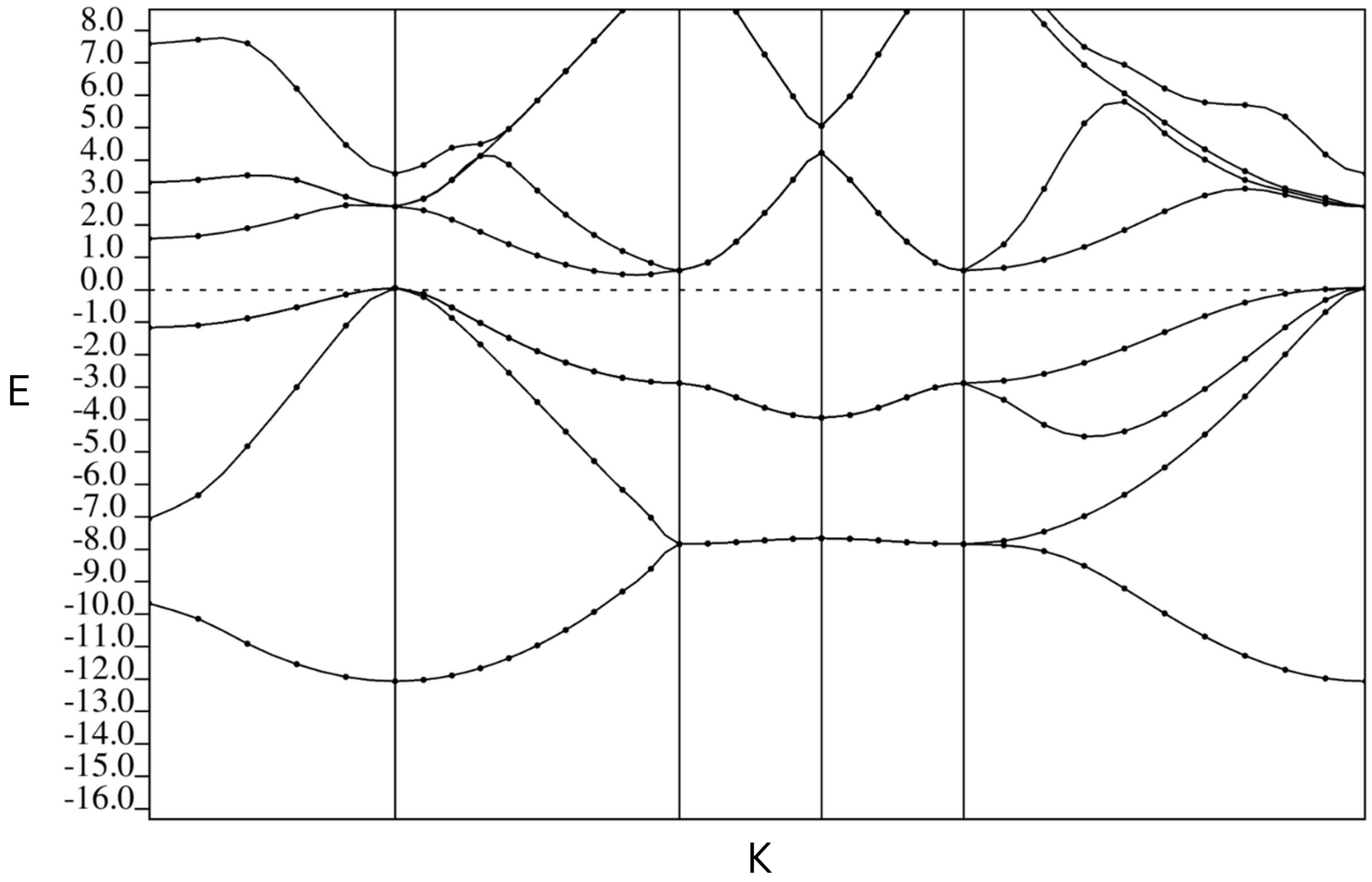
# Schrödinger equation

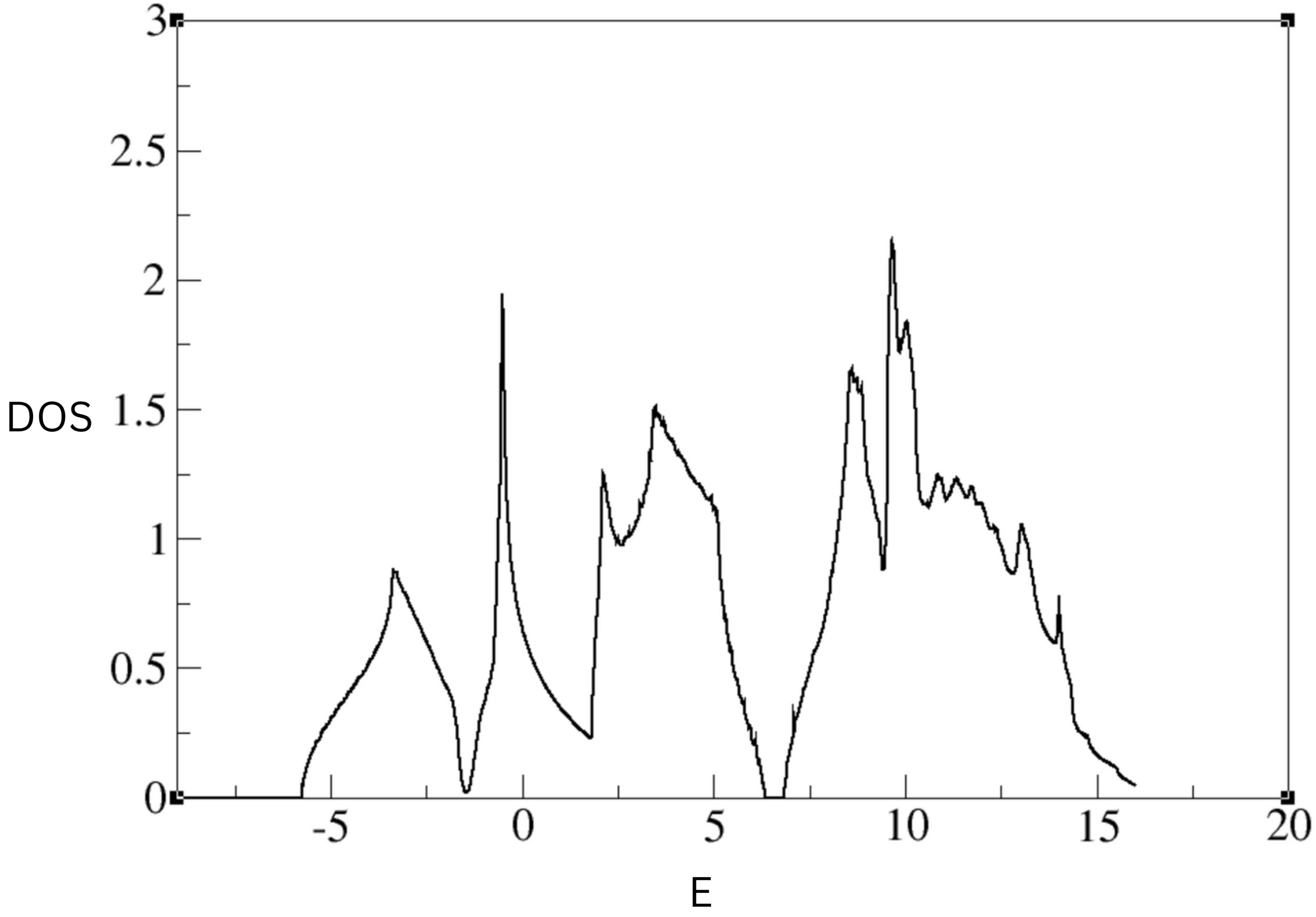
$$\hat{\mathrm{H}}\,|\Psi\rangle = E|\Psi\rangle$$

$$\left[\quad \frac{h^2}{2m}\sum_{i\quad 1}^{N}\nabla_i^2 + \sum_{i\quad 1}^{N} V(\mathbf{r}_i) + \sum_{i\quad 1}^{N}\sum_{j<i} U(\mathbf{r}_i,\, \mathbf{r}_j)\right]\psi = E\psi.$$

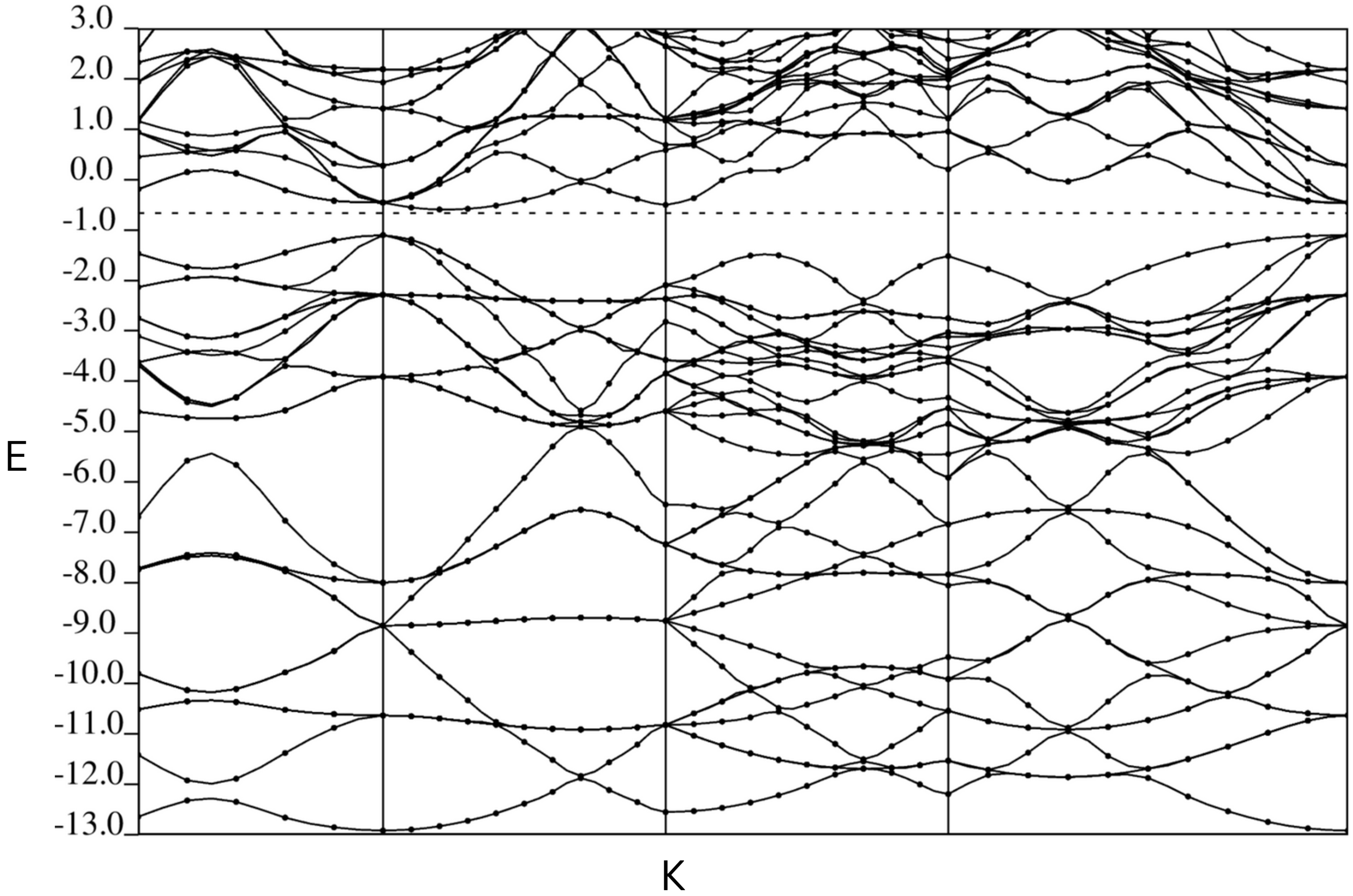
# Band Structure (Si)



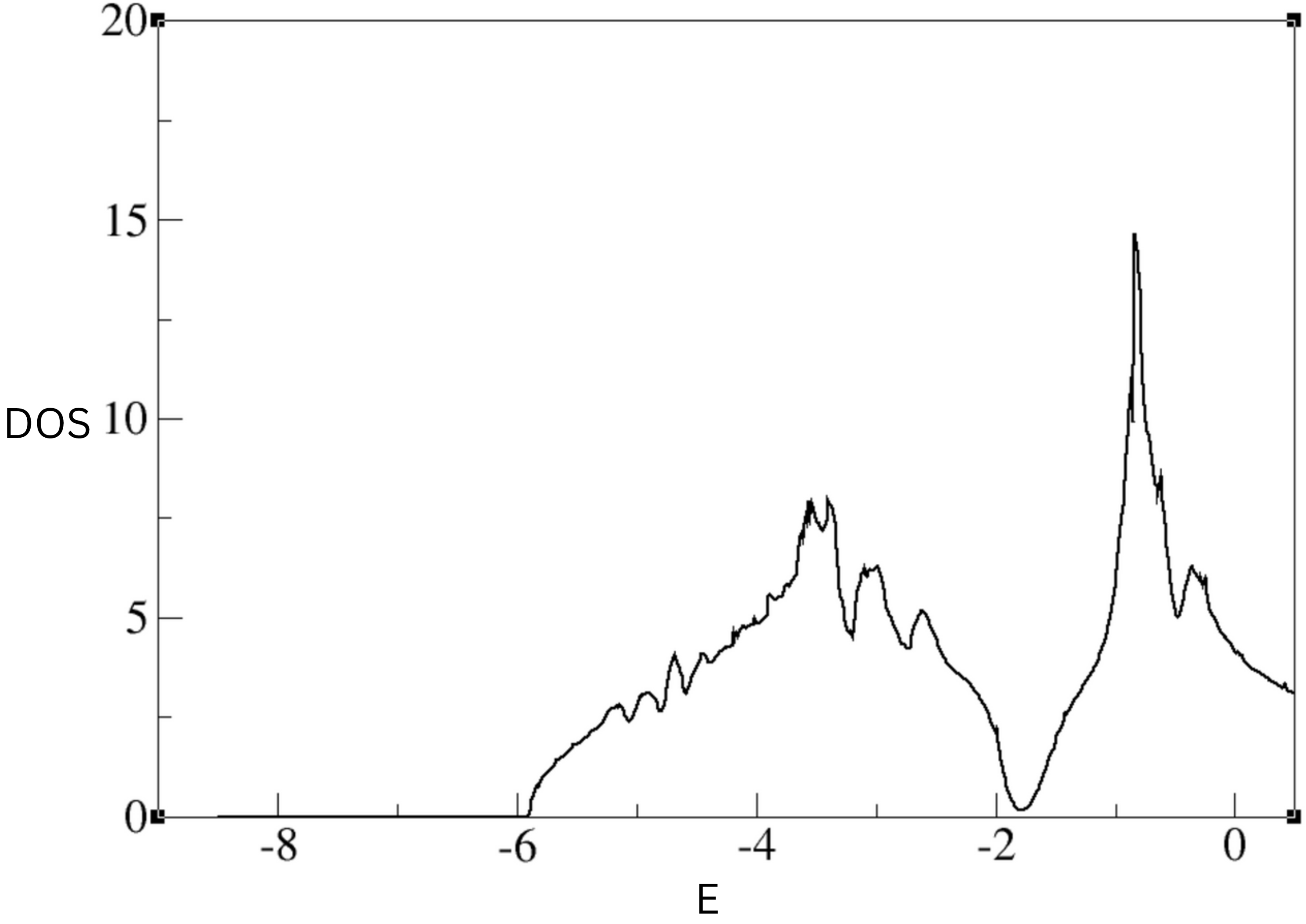
# Density of States (DOS)



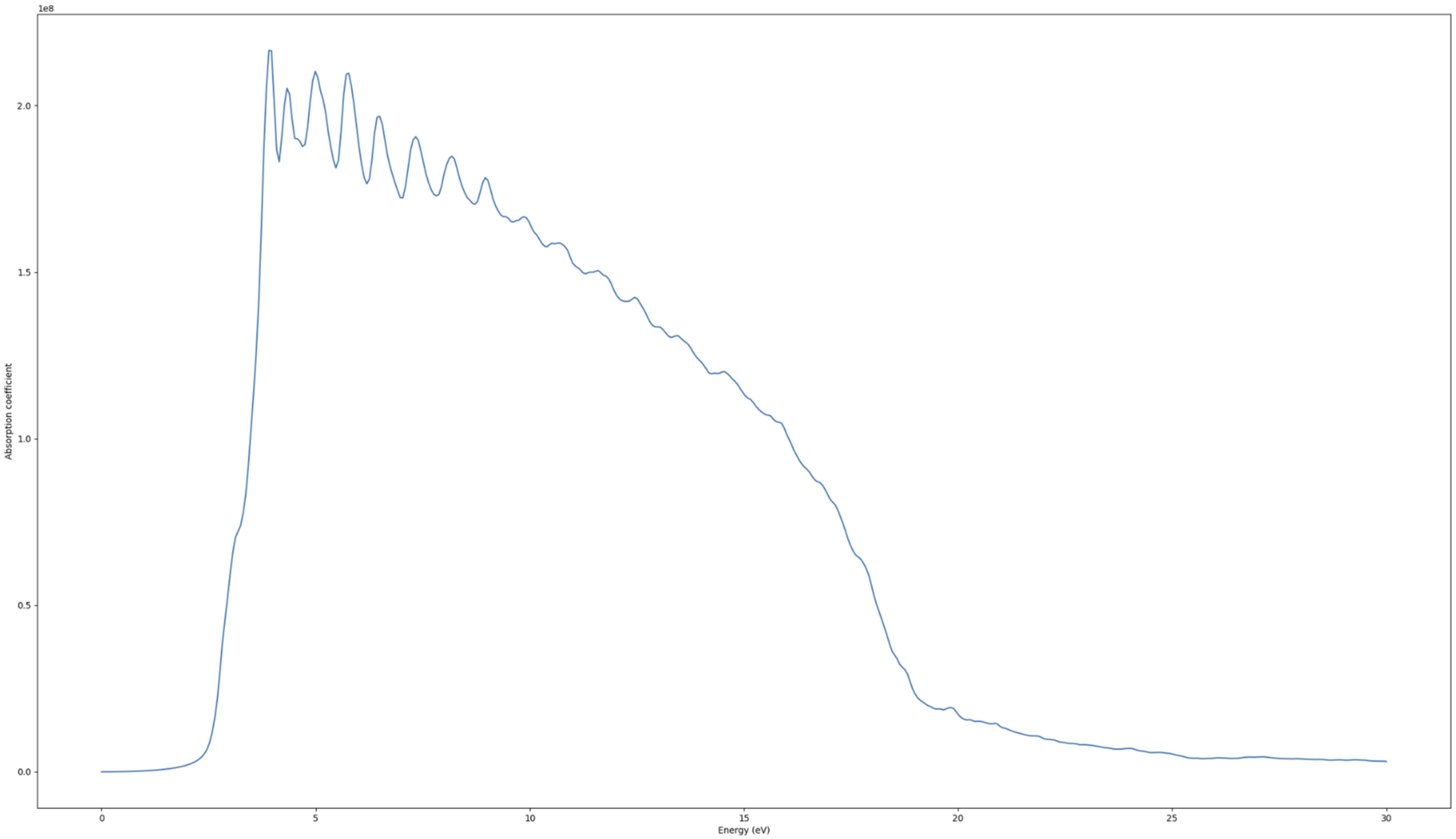
# Band Structure (Si)



# Density of States (DOS)

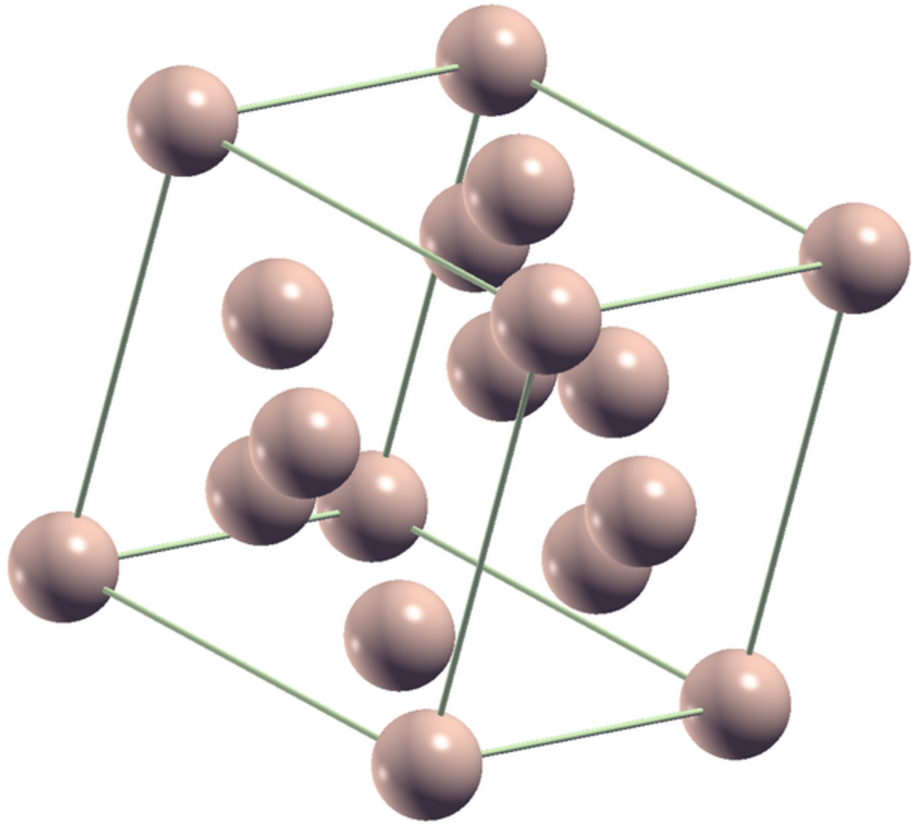


# Absorption Coefficient



# Cohesive energy

Cohesive energy = E bulk - E isolated



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   prefix='silicon',
   tstress = .true.
   tprnfor = .true.
   pseudo_dir = './',
   outdir='./'
/
 &system
   ibrav= 1, celldm(1)=50.00, nat= 1 , ntyp= 1,
   ecutwfc =35.0,occupations = "smearing",degauss = 0.001

/
 &electrons
   mixing_mode = 'plain'
   mixing_beta = 0.7
   conv_thr =  1.0d-8
/
ATOMIC_SPECIES
 Si  28.086  Si.pbesol-n-rrkjus_psl.1.0.0.UPF
ATOMIC_POSITIONS alat
 Si 0.00 0.00 0.00
K_POINTS gamma
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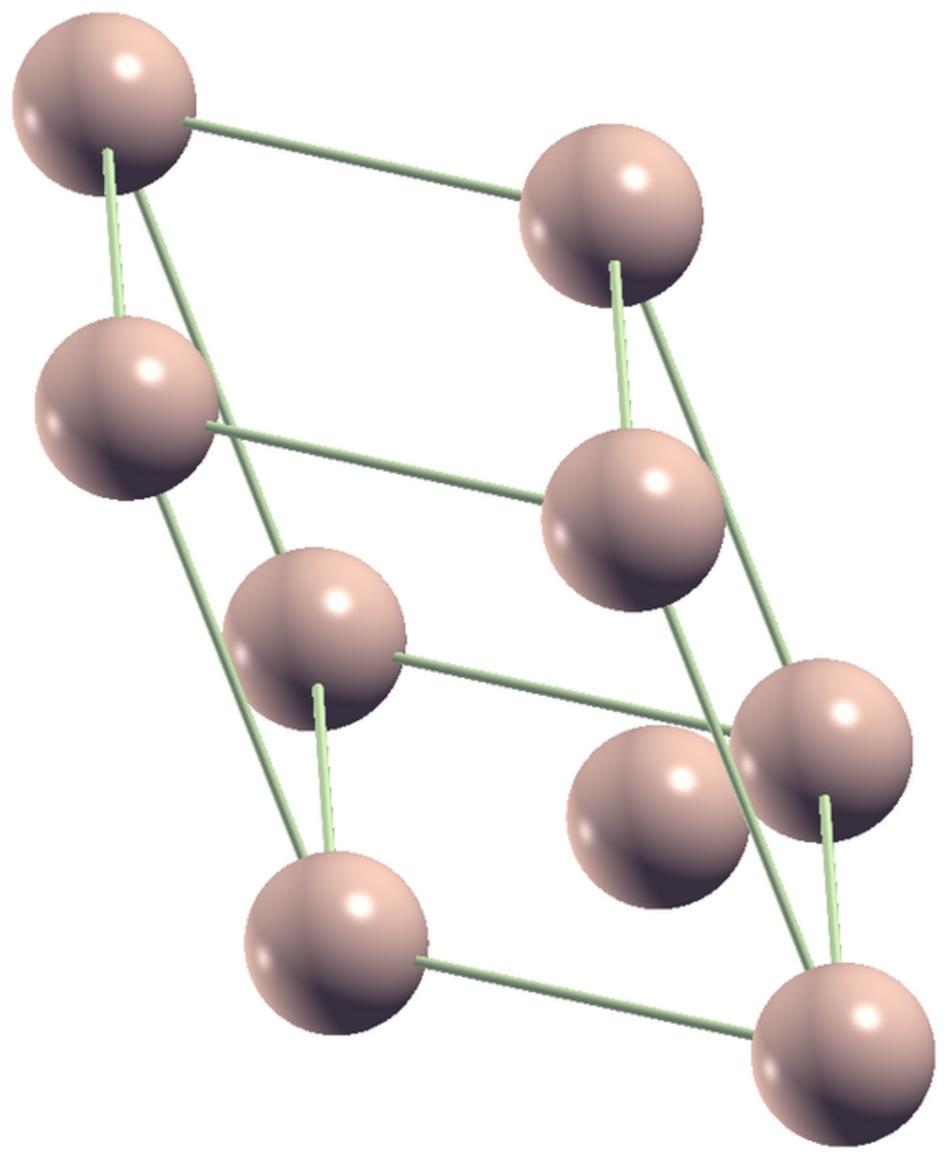
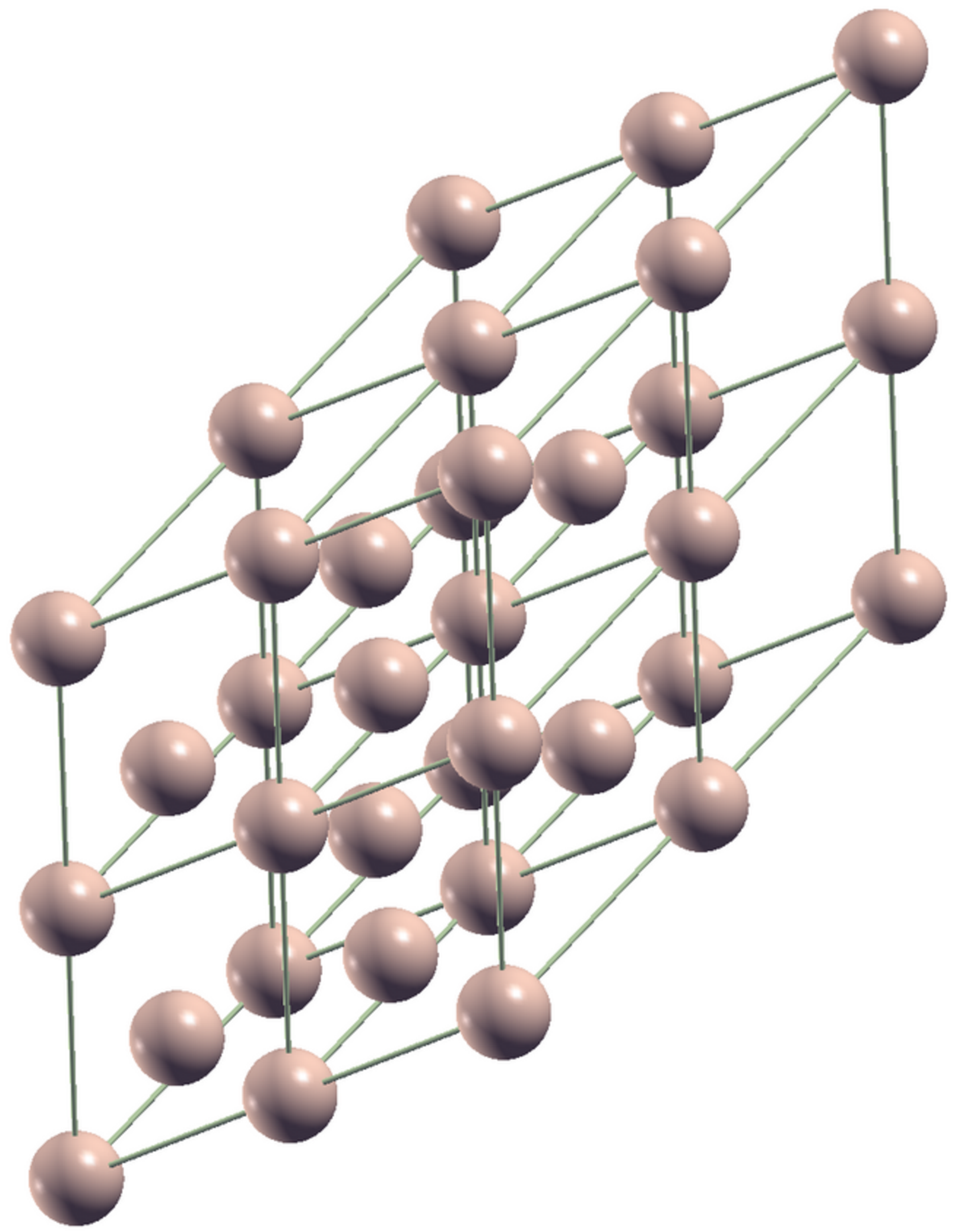
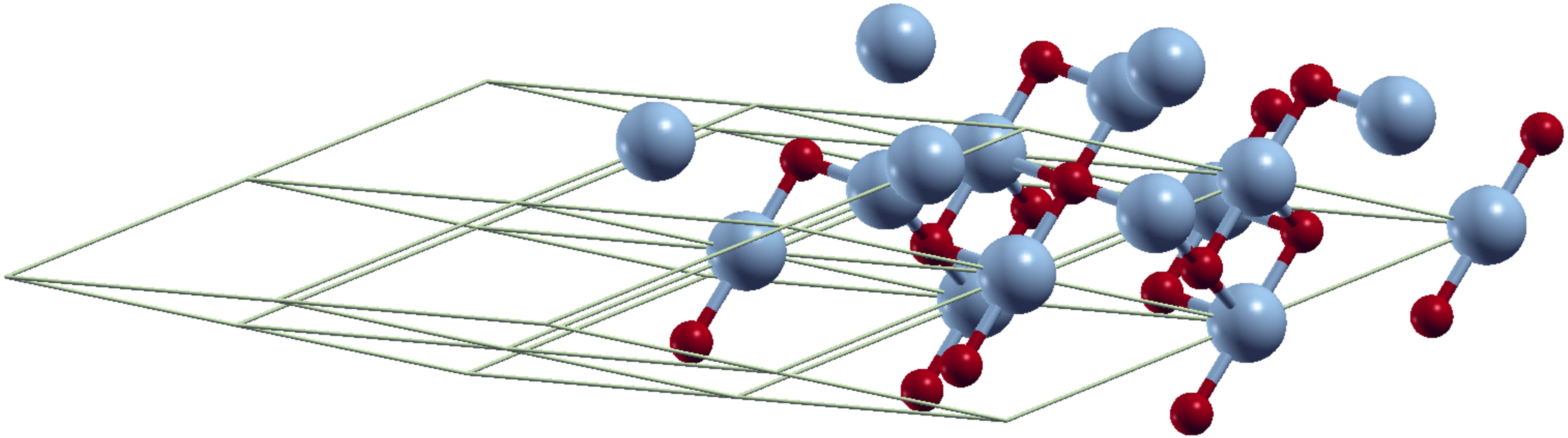
# Defect Formation Energy

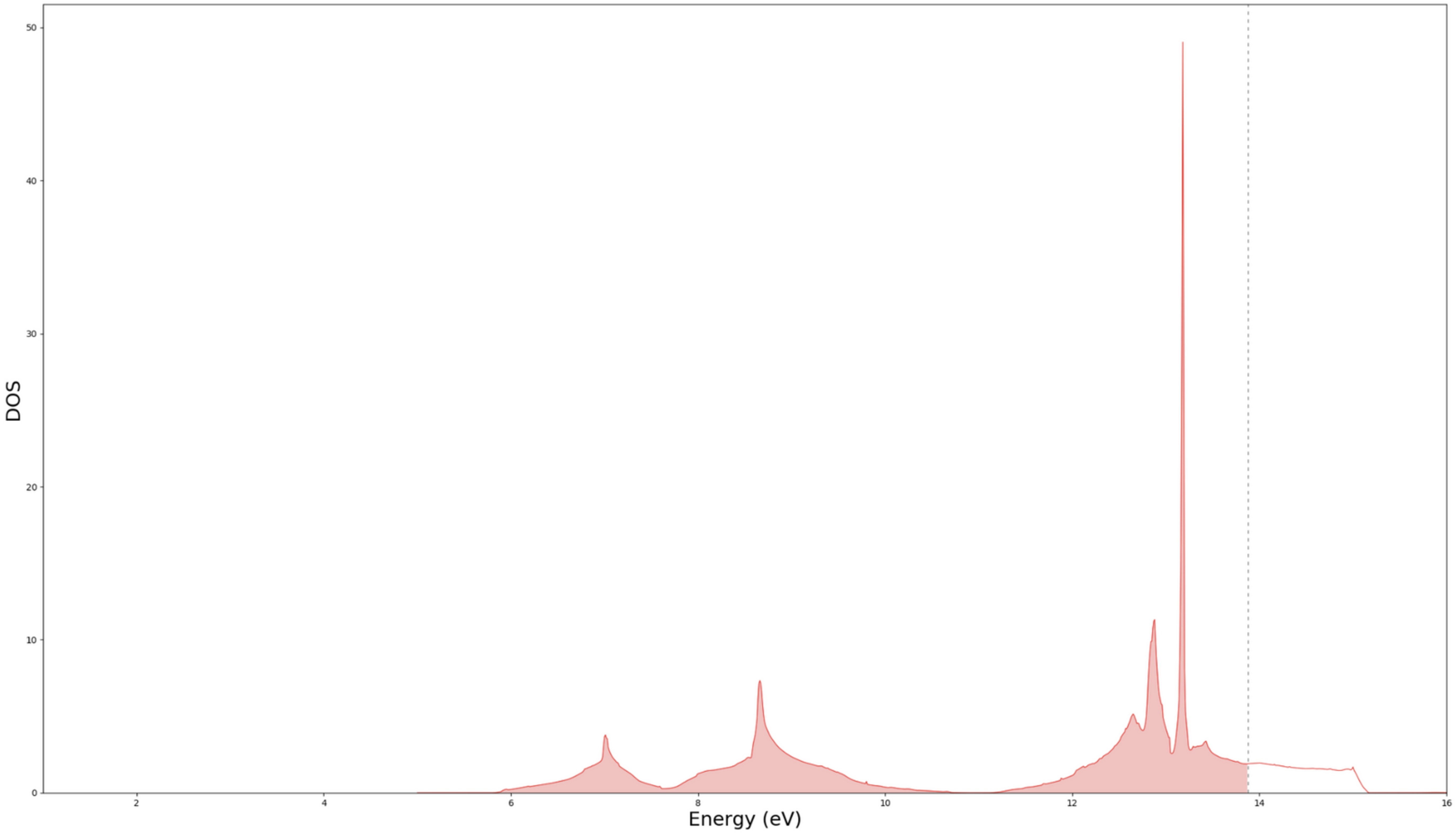




# NiO



# NiO (DFT)



# NiO (DFT + U)